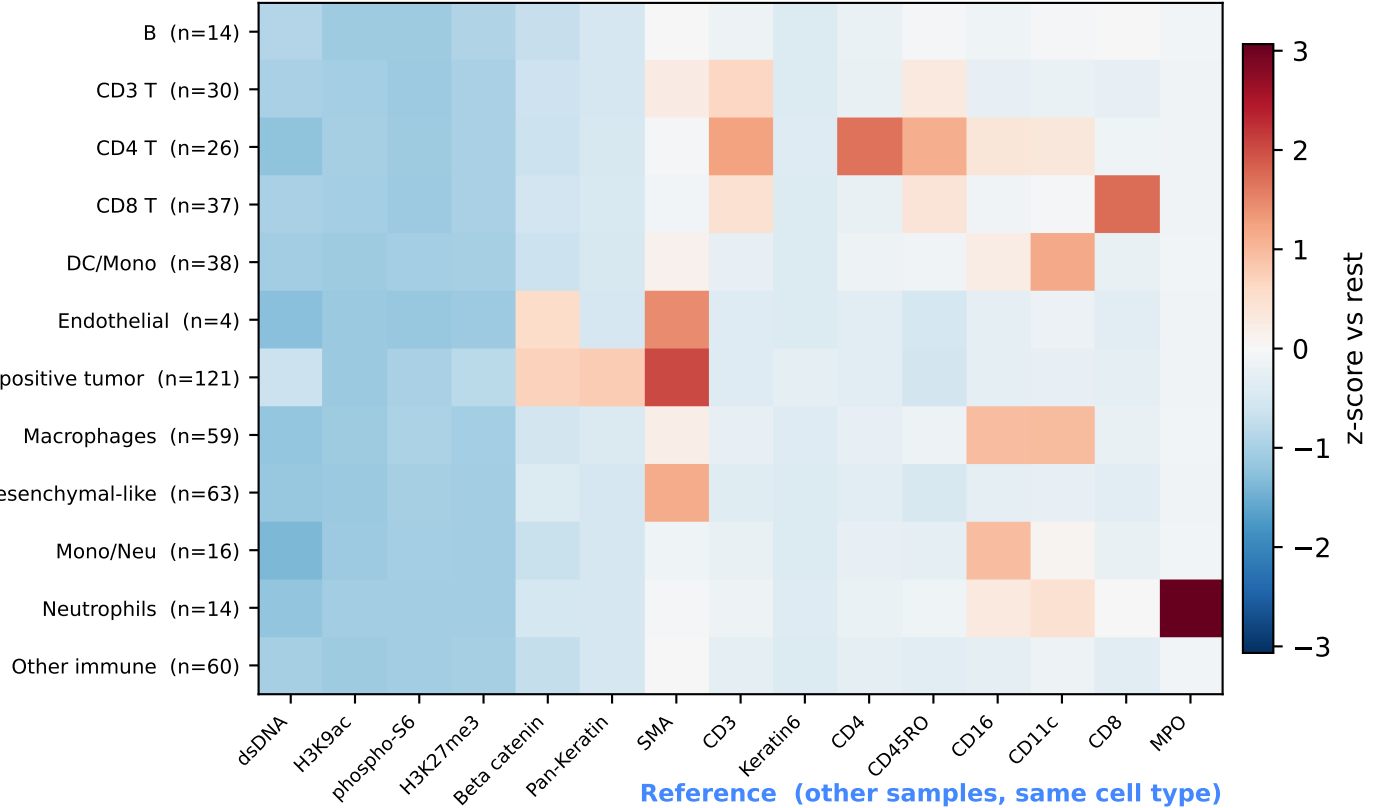
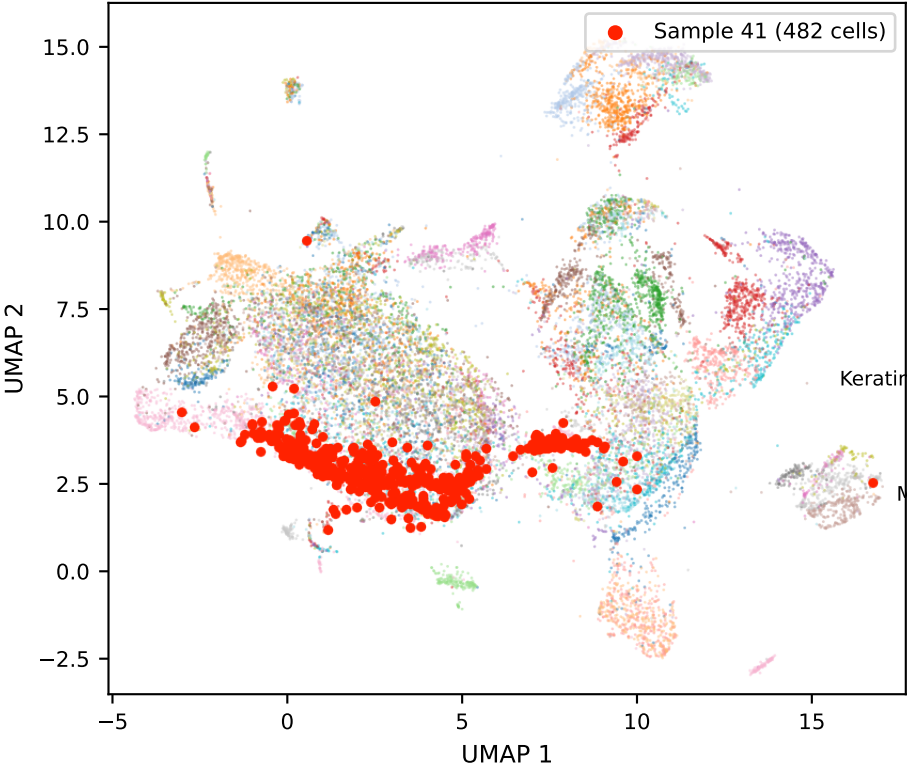
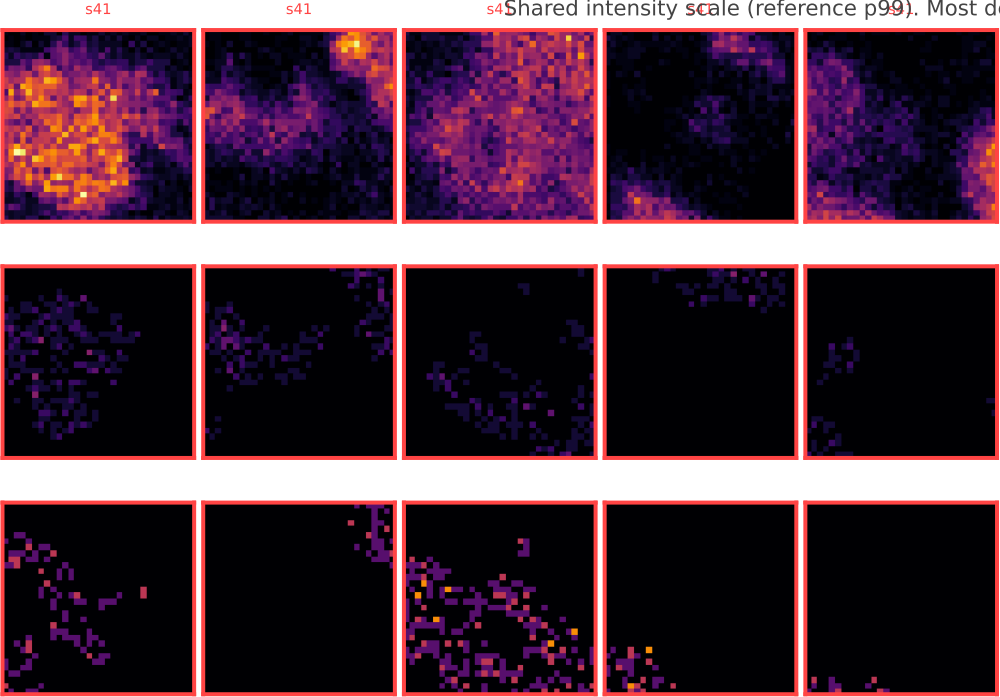


Sample 41 batch effect diagnosis — MIBI_TNBC_CIM_VICReg_Funnel_Large
All cell types show globally depressed nuclear/proliferation markers → technical staining quality issue
UMAP — sample 41 highlighted in red
Sample 41 marker expression — z-score vs all other samples
(consistent pattern across cell types = technical batch effect)



Sample 41 (Keratin-positive tumor)



Reference (other samples, same cell type)

